

Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

AI Library Assistant Agent | Library & Book Management Domain

Detail	Description
Audience	Engineering & Arts students (challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — mirrors reference career-advisor pattern

1. Problem Statement

The Rajalakshmi Engineering College central library receives a continuous stream of student emails and portal requests related to book borrowing, renewals, reservations, and overdue fine queries. Currently, library staff manually read every request, identify the book and student details, check due dates, calculate applicable fines, and decide whether to renew, reserve, or flag the item as overdue. This manual process is repetitive, slows down during peak semester periods (assignment submissions and exam preparation), and leads to inconsistent fine calculation and delayed renewals.

Design and build an Autonomous UiPath Agent, the AI Library Assistant Agent, that reads an incoming student library request, reasons over its content, classifies it, extracts structured details, and independently completes the required action end-to-end — without any human approval step — by invoking RPA workflows and Generative AI activities as tools.

2. System Prompt

You are an AI Library Assistant Agent for Rajalakshmi Engineering College.
Your goal is to analyze student library requests and decide the best possible action.

Follow this reasoning process:

1. Read and understand the request content.
2. Classify it into one of the following:
 - Renewal Request
 - Reservation Request
 - Overdue Fine Query
 - Other
3. If it is Renewal, Reservation, or Overdue Fine Query:
 - Extract student register number, book title/ISBN, and due date
 - Decide if it is urgent (e.g., due date within 2 days or already overdue)
 - If urgent, recommend immediate logging and notification
 - Generate a professional acknowledgement reply
4. If it is Other:
 - Skip processing

Rules:

- Think step-by-step before answering
- Be concise and structured
- Do not assume missing details
- Return output in structured format

3. User Prompt

```
Analyze the following student library request and decide the appropriate action:
{{request_text}}
```

4. Input Schema

```
{
  "request_text": "string"
}
```

5. Output Schema

```
{
  "category": "string",
  "register_number": "string",
  "book_title": "string",
  "due_date": "string",
  "urgency": "string",
  "action": "string"
}
```

6. Tool: Build Workflow & Publish

Create the following process inputs for the RPA workflow the agent will invoke:

- category (text)
- register_number (text)
- book_title (text)
- due_date (text)
- urgency (text)

7. Add Tool: Map Inputs

Agent Field	Process Argument
category	in_category
register_number	in_register_number
book_title	in_book_title
due_date	in_due_date
urgency	in_urgency

8. Updated System Prompt — Tool Invocation Logic

```
If the request is classified as Renewal Request, Reservation Request,
or Overdue Fine Query:
  - Use the log_to_excel tool to store the extracted student and book
    details in the library transaction register.
```

- Use the generate_reply GenAI activity to draft a professional acknowledgement email to the student.
- If urgency = "High", use the notify_library_desk RPA workflow to alert the library front-desk staff automatically (fully autonomous, no manual approval required).

If the request is classified as Other:

- Take no further action.

9. Tools & Activities Summary

Tool / Activity	Type
log_to_excel	RPA Workflow (UiPath Excel Automation)
generate_reply	GenAI Activity (LLM prompt-based text generation)
notify_library_desk	RPA Workflow (Outlook / Teams notification)
classify_request	GenAI Activity (embedded in Agent reasoning)

Note: This use case is intentionally fully autonomous — the agent completes classification, extraction, logging, notification, and reply generation without an escalation or human-review step, demonstrating end-to-end autonomous decision-making for the workshop challenge.

10. Expected Learning Outcome

- Design a multi-step reasoning system prompt for an autonomous agent
- Define structured input/output schemas for agent-to-workflow communication
- Map agent-extracted fields to RPA process arguments
- Combine RPA workflows and GenAI activities as callable tools within one agent
- Publish and test a complete autonomous agent in UiPath Agent Builder